

Modalities:

Top choices:

- Regulatory regions
- More proteins

DNA sequence:

- One-hot array
- Ref-alt encoding
- Foundation model embedding
- K-mers (31 in this case)
 - Not ideal for TB
- Explicitly separated regions
 - E.g. passing regulatory non-coding regions as separate modality to protein regions
 - Interesting approach but we would need to retain information about what non-coding and coding regions are adjacent

Regulatory regions:

- Refer to datasets provided in email to get specific features that relate specifically to regulatory regions and their relationships
 - Particularly in the regime we are already working
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Protein Sequence:

- One-hot array
- Ref-alt encoding
- Embedding
- *Its possible we could get some gains by simply looking at more genes in the sequence than just the ones of note*
 - Could be done within Big-TB

3D structure:

- Distance matrix (e.g. alpha fold)
 - Would only do this for proteins that form a physical complex in the cell
 - Does not make a lot of sense if they are not touching
- Density vector (mine)
- Shrake-Rupley algorithm
 - Not a bad idea but probably overkill for this particular task as the information this gives us we already have an in better methods than this approach
- Useful to encode if two proteins do in fact interact
 - EMB CA&B RPOB&C are known to interact (in genetic order)
 - This could be a 0-1 but that is a bit too rough

Molecular Characteristics:

- Molecular weight (g/mol)
- Isoelectric point
- Hydrophobicity
- Lipophilicity
 - Kinda inverse of hydrophobicity
 - In reality they are not because they are calculated in different ways by different people
- Refractivity
 - Refers to polar surface area, related to above but not entirely dependent

This can be gained from using RDKit:

- This is probably not ideal as its too much information
- *Not recommended*

Lineage:

- 1D binary vector
- N X N matrix

Other Features:

- Unitigs
 - “Unambiguous contigs”
 - Derived contigs done without reference genome
 - Basically DNA but derived more on the fly
 - Should handle complex clone errors better
- Graphical representation
 - SNP distance
 - Genetic distance between two samples from SNPs between them
 - FCGR distance
 - Frequency chaos game representation.
 - Converts to a 2D image based on k-mer frequency
 - Handles reversals or other errors better

Potential model designs:

Asymmetrical Cross-attention:

Summary:

Baseline performance shows strong 1D CNN scores with protein and other models picking up important but secondary features. The model would use cross-attention blocks treating the DNA

sequence as Query with other modalities as Key / Value. Should allow 1D CNN insights to guide inference only retrieving information from other modalities when needed (in theory).

Best for resolving ambiguous mutations by querying a “more robust” biological context

Inputs:

1-Hot DNA sequence

Options:

- Any, cross attention allows high flexibility

Late-stage routing with modality dropout or frozen train:

Summary: it's difficult to fuse highly disparate modalities as it cannot be done spatially / token-for-token. Inference is done with three branches that are fused near the end of the model. We then drop out modalities to ensure a robust model.

Alternatively we could freeze modules, e.g. train in 3 stages: train DNA and protein encoders on just sequences, freeze them, add in the lineage branches at the end of the model concat with frozen latents. Only train final dense heads off this at this stage.

Best for fusing highly heterogeneous data with different dimensions particularly if they cannot be spatially aligned.

Branches:

- Local sequence features
 - k-mers
 - Handles specific genomic fragments and their frequency without needing full alignment
- Global graph features
 - UNitigs and FCGR or SNP distance
 - Uses species-agnostic signals and features
 - Requires GCN
- Population context
 - Lineage information
 - This branch will have random drop-out to ensure they do not over use this modality (as has been observed)

Potential Model Choices (not entire model)

Adversarial lineage decoupling:

- **Effect: train model to to over rely on lineage as some found this could be used to shortcut the inference**

- Modeled after unlearning to have a model (doesn't matter what) train to predict ABR from WGS data.
- Separate head predicts lineage
 - Gradient for this head is inverted during back propagation to actively penalize lineage based inference ideally forcing the latent space to be lineage agnostic

Predict BOTH MIC and ABR:

- **Effect: Forces the model to “understand” both.**
- In theory this could force the model latent space to organize around biological severity of a mutation not just a binary while retaining binary classification
- Same intuition behind a number of similar methods.
- *Trade off is we often do not have MIC values, check this*
 - Its qualitative (binned)

Causal Interpretability via Probing:

- Effect: won't improve current model but may provide actionable insights into model design or simply interpretable features.
- We can maintain a sparse ref-alt embedding for a set of canonical variants.
- Inject these latents during inference to observe how prediction changes based on local specific interventions
- In general mech interp is not very applied in this space:
 - SOTA is SHAP
 - Models tend to find confounders and the most common confounder they find is resistance to other antibiotics.
 - Often the best indicator if 2nd line drug resistance is if they have first line resistance
 - Have to be very careful to find actual causal relationships not just cofounded correlations

Meeting Notes:

- In practice transformers have been less successful than CNN
 - Likely because we lack datapoints and importantly unique datapoints
- Big picture we are mostly focused on interp not performance